SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: **Database Management Systems**
COURSE CODE: **CSA05**

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Short Answer Text(High)
Assessment Tool Weightage: 30%

Dr

QUESTIONS		Total Marks: 50	
Q.No	**Question**	**Bloom's Level**	**Marks**
1	UPS Product Tracking Information System UPS prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, UPS relies on a company-wide information system. Shipped items are the heart of the UPS product tracking information system. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date. Shipped items are received into the UPS system at a single retail center. Retail centers are characterized by their type, uniqueID, and address. Shipped items make their way to their destination via one or more standard UPS transportation events (i.e., flights, truck deliveries). These transportation events are	BL1 – Remember	5

	characterized by a unique scheduleNumber, a type (e.g, flight, truck), and a deliveryRoute. Create an Entity Relationship diagram that captures this information about the UPS system.		
2	Course Management System Consider a university database for the scheduling of classrooms for final exams. This database could be modeled as the single entity set exam, with attributes course-name, section-number, room-number, and time. Alternatively, one or more additional entity sets could be defined, along with relationship sets to replace some of the attributes of the exam entity set, as • course with attributes name, department, and c-number • section with attributes s-number and enrollment, and dependent as a weak entity set on course • room with attributes r-number, capacity, and building Show an E-R diagram illustrating the use of all three additional entity sets listed.	BL2 – Understand	5
3	University Registrar's Office Management System A university registrar's office maintains data about the following entities: (a) courses, including number, title, credits, syllabus, and prerequisites; (b) course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom; (c) students, including student-id, name, and program; (d) instructors, including identification number, name, department, and title. (e) Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled. Construct an E-R diagram for the registrar's office.	BL2 – Understand	5

4	Hospital Management System Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted. Construct appropriate tables for the above ER Diagram : Patient(SS#, name, insurance) Physician (name, specialization) Test-log(SS#, test-name, date, time) Doctor-patient (physician-name, SS#) Patient-history(SS#, test-name, date)	BL2 – Understand	5
5	Company Project Management System Construct an ER Diagram for Company having following details : Company organized into DEPARTMENT. Each department has unique name and a particular employee who manages the department. Start date for the manager is recorded. Department may have several locations. 🎬 A department controls a number of PROJECT. Projects have a unique name, number and a single location. 🎬 Company's EMPLOYEE name, ssno, address, salary, sex and birth date are recorded. An employee is assigned to one department, but may work for several projects (not necessarily controlled by dept). Number of hours/week an employee works on each project is recorded; The immediate supervisor for the employee. 🎬 Employee's DEPENDENT are tracked for health insurance purposes (dependentname, birthdate, relationship to employee).	BL2 – Understand	5
6	Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.	BL1 – Remember	5
7	Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.	BL2 – Understand	5

8	Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.	BL2 – Understand	5
9	Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.	BL2 – Understand	5
10	Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.	BL1 – Remember	5

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Concept Identification	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Linkages	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Organization	20%	Correctly applies concepts to solve the	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application

		question/proble m			
Integration of Literature	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity	10%	Clear, concise, and well-organized answer	Understandab l e with minor issues	Somewhat unclear or poorly structured	Difficult to understand

CO1-ASSESSMENT TOOL-1-(CSA0508_C01_AT1_SHORT ANSWER)

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COURSE CODE: CSA0508
COURSE NAME:DATABASE MANAGEMENT SYSTEM
DATE:21.07.2026

Question 1: UPS Product Tracking Information System

Entities and Attributes

1. Shipped Item

- Item_Number (**Primary Key**)
- Weight
- Dimensions
- Insurance_Amount
- Destination
- Final_Delivery_Date

2. Retail Center

- Center_ID (**Primary Key**)
- Type
- Address

3. Transportation Event

- Schedule_Number (**Primary Key**)
- Type (Flight/Truck)
- Delivery_Route

Relationships

- A **Retail Center** receives many **Shipped Items**. (**1 : M**)
- A **Shipped Item** travels through one or more **Transportation Events**. (**M : N**)

ER Diagram

```
+--------------+
| Retail Center|
+--------------+
        |
    Receives
```

```
        |
        |
+-------------+
| Shipped Item |
+-------------+
        |
    Travels By
        |
+--------------------+
| Transportation Event|
+--------------------+
```

Explanation

The Shipped Item is the main entity in the UPS tracking system. Each shipped item has a unique item number and contains details such as weight, dimensions, insurance amount, destination, and final delivery date. Every shipped item is received at one Retail Center, which has a unique center ID, type, and address. During delivery, the shipped item passes through one or more Transportation Events, such as flights or truck deliveries. Each transportation event has a unique schedule number, type, and delivery route. This ER diagram helps UPS efficiently track every package from the retail center to its final destination.

2. Course Management System

Entities and Attributes

1. Course

- Course_Name (**Primary Key**)
- Department
- Course_Number

2. Section (Weak Entity)

- Section_Number
- Enrollment

3. Room

- Room_Number (**Primary Key**)

- Capacity
- Building

4. Exam

- Time

Relationships

1. A **Course** has many **Sections**.
    - Relationship: **Has**
    - Cardinality: **1 : M**
2. A **Section** has an **Exam**.
    - Relationship: **Scheduled**
3. An **Exam** is conducted in a **Room**.
    - Relationship: **Held In**

ER Diagram

```
+-----------+
|  Course   |
+-----------+
| Course_No |
| Name      |
| Dept      |
+-----------+
      |
     Has
      |
      |
+------------+
|  Section   |
+------------+
| Section_No |
```

```
| Enrollment |
+------------+
      |
   Exam
      |
+-----------+
|   Room    |
+-----------+
| Room_No   |
| Capacity  |
| Building  |
+-----------+
```

Explanation

The **Course Management System** stores information about courses, sections, exams, and rooms. A course can have multiple sections. Each section has an exam scheduled at a particular time. The exam is conducted in a specific room. The section entity is considered a weak entity because it depends on the course entity for its existence.

This ER diagram helps the university manage final exam schedules efficiently.

3. University Registrar's Office Management System

Explanation

A University Registrar's Office Management System stores information about courses, students, instructors, course offerings, and student enrollments. It also records the grades obtained by students in each course.

Entities

1. Course

Stores details of each course.

Attributes

- Course_ID (Primary Key)
- Title
- Credits
- Syllabus

- Prerequisite

2. Course Offering

Represents a particular offering of a course in a semester.

Attributes

- Offering_ID (Primary Key)
- Year
- Semester
- Section
- Timing
- Classroom

3. Student

Attributes

- Student_ID (Primary Key)
- Name
- Program

4. Instructor

Attributes

- Instructor_ID (Primary Key)
- Name
- Department
- Title

5. Enrollment

Represents the relationship between Student and Course Offering.

- Grade

ER diagram

```
        COURSE
    ---------------
    Course_ID (PK)
    Title
```

```
           |
         1 | M
           |
    COURSE_OFFERING
    -----------------
    Offering_ID (PK)
    Semester
    Section
           |
     ----------------
     |              |
   M |              | 1
     |              |
 STUDENT       INSTRUCTOR
------------   -------------
Student_ID     Instructor_ID
Name           Name
     |
   M | N
     |
 ENROLLMENT
------------
Grade
```

Relationships

- Course → Course_Offering = 1 : M
- Course_Offering → Instructor = M : 1
- Student ↔ Course_Offering = M : N (through Enrollment)
- Enrollment stores the Grade.

4.Hospital Management System

Explanation

A Hospital Management System maintains information about patients, physicians, medical tests, and patient history. It stores patient details, doctor details, tests conducted on patients, and previous medical records.

Entities and Attributes

1. PATIENT

Stores details of patients.

Attributes:

- **SS# (Primary Key)**
- Name
- Insurance

2. PHYSICIAN

Stores details of doctors.

Attributes:

- **Physician_Name (Primary Key)**
- Specialization

3. TEST_LOG

Stores details of tests performed on patients.

Attributes:

- SS# (Foreign Key)
- Test_Name
- Date
- Time

4. PATIENT_HISTORY

Stores previous test records of patients.

Attributes:

- SS# (Foreign Key)
- Test_Name
- Date

ER Diagram

```
          PATIENT
     ----------------
     SS# (PK)
     Name
     Insurance
          |
  Undergoes
          |
          M
          |
     TEST_LOG
  -----------------
  Test_Name
  Date
  Time


PATIENT ---------------------- PHYSICIAN
   M         Treated By          M


     PHYSICIAN
  -----------------
  Name (PK)
  Specialization
```

```
PATIENT
   |
   | Has
   |
   M
PATIENT_HISTORY
-----------------------
Test_Name
Date
```

Relationships

- Patient → Test_Log : One patient can have many test records (1:M).
- Patient ↔ Physician : A patient can be treated by many physicians, and a physician can treat many patients (M:N).
- Patient → Patient_History : One patient can have many history records (1:M).

5: Company Project Management System

Explanation

A company database stores details about departments, employees, projects, and dependents. It manages employee assignments, project details, and department information.

Entities and Attributes

1. DEPARTMENT

- Dept_ID (PK)
- Dept_Name
- Location
- Manager

2. EMPLOYEE

- SSN (PK)
- Name
- Address

- Salary
- Sex
- Birth_Date

3. PROJECT

- Project_ID (PK)
- Project_Name
- Location

4. DEPENDENT

- Dependent_Name
- Birth_Date
- Relationship

ER Diagram

```
     DEPARTMENT

       ----------------

       Dept_ID (PK)

       Dept_Name

       Location

       Manager

             |

             | 1:M

             |

          EMPLOYEE

       ----------------

       SSN (PK)

       Name

       Address

       Salary

       Sex

       Birth_Date
```

```
          /       \
         /         \
      1:M           M:N
       /              \
DEPENDENT           PROJECT
-----------        -------------
Dep_Name           Project_ID(PK)
Birth_Date         Project_Name
Relationship       Location
```

Relationships

1. **Department - Employee (1:M)**
   - **One department has many employees.**
2. **Department - Project (1:M)**
   - **One department controls many projects.**
3. **Employee - Project (M:N)**
   - **Employees can work on many projects.**
4. **Employee - Dependent (1:M)**
   - **One employee can have many dependents**

6: Explain Schema and Instance with Examples. Distinguish between Logical Schema, Physical Schema, and View Schema.

1. Schema

Definition:

A schema is the structure or design of a database. It defines how data is organized, including tables, attributes, relationships, and constraints.

A schema is created during the database design stage and does not change frequently.

Example:

Student Database Schema

```
STUDENT
```

```
------------------------
Student_ID (Primary Key)
Name
Department
Age
```

This schema defines the structure of the Student table.

2. Instance

Definition:

An **instance** is the actual collection of data stored in a database at a particular point of time.

The instance changes whenever data is inserted, updated, or deleted.

Example:

Student Table Instance:

Student_ID	**Name**	**Department**	**Age**
101	Arun	CSE	20
102	Priya	IT	21
103	Rahul	ECE	19

Here, the records are called the instance of the database.

Difference Between Schema and Instance

Schema	Instance
It is the design/structure of the database	It is the actual data stored in the database.
Created during database design.	Created when users insert data.
Changes rarely.	Changes frequently.
Example: Table structure.	Example: Table records

Types of Schema

1. Logical Schema

Definition:

Logical schema describes the logical structure of the database. It defines tables, attributes, keys, and relationships without considering physical storage.

Example:

```
STUDENT
```

```
------------------
Student_ID
Name
Department

COURSE
------------------
Course_ID
Course_Name
Credits
```

Features:

- Describes database design.
- Used by database designers.
- Independent of hardware.

2. Physical Schema

Definition:

Physical schema describes how the data is actually stored in the computer system.

It deals with storage details like files, indexes, and memory organization.

Example:

- Student table stored in database files.
- Index created on Student_ID for faster searching.

Features:

- Related to storage techniques.
- Managed by DBA.
- Improves database performance.

3. View Schema

Definition:

View schema describes the part of the database visible to a particular user.

Different users can have different views of the same database.

Features:

- Provides data security.
- Shows only required information.
- Helps different users access different views.

Difference Between Logical, Physical, and View Schema

Logical Schema	**Physical Schema**	**View Schema**
Describes database structure	Describes data storage	Describes user view
Includes tables and relationships	Includes files and indexes	Includes selected data
Used by designers	Used by DBA	Used by users
Independent of storage	Depends on hardware	Depends on user requirements

Conclusion

A database schema defines the structure of a database, while an instance represents the actual data stored at a specific time. Logical schema explains the database design, physical schema explains storage details, and view schema provides customized views for users.

7: Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.

Explanation

A Bank Management System manages information about customers, accounts, branches, and loans. It stores customer details, account details, and loan information.

Entities and Attributes

1. CUSTOMER

Stores customer information.

Attributes:

- **Customer_ID (Primary Key)**

- Name
- Address
- Phone_No

2. ACCOUNT

Stores bank account details.

Attributes:

- **Account_No (Primary Key)**
- Account_Type
- Balance

3. BRANCH

Stores branch information.

Attributes:

- **Branch_ID (Primary Key)**
- Branch_Name
- Location

4. LOAN

Stores loan details.

Attributes:

- **Loan_ID (Primary Key)**
- Loan_Type
- Amount

Relationships

1. **Customer – Account (M:N)**
    - A customer can have multiple accounts.
    - An account can belong to multiple customers (joint account).

2. **Branch – Account (1:M)**
    - One branch maintains many accounts.
    - Each account belongs to one branch.
3. **Customer – Loan (M:N)**
    - A customer can take multiple loans.
    - A loan can be taken by multiple customers.

ER Diagram

```
               CUSTOMER

           ----------------

           Customer_ID(PK)

           Name

           Address

           Phone_No

              /       \

             /         \

           M:N         M:N

           /              \

      ACCOUNT            LOAN

    -------------      -------------

    Account_No(PK)     Loan_ID(PK)

    Type               Loan_Type

    Balance            Amount

         |

         |

        M:1

         |

         |

       BRANCH

    -------------
```

```
Branch_ID(PK)

Branch_Name

Location
```

8: Explain the Extended ER (EER) Model. Describe the concepts of Generalization, Specialization, and Aggregation.

Extended ER (EER) Model

Definition:

The Extended Entity Relationship (EER) Model is an advanced version of the ER model. It includes additional concepts like inheritance, specialization, generalization, and aggregation to represent complex real-world systems.

The EER model helps in designing databases with more detailed relationships between entities.

Concepts of EER Model

1. Generalization

Definition:

Generalization is the process of combining two or more lower-level entities into a single higher-level entity based on common attributes.

It follows a **bottom-up approach**.

Example:

```
STUDENT           TEACHER
       \              /
        \            /
         \          /
          PERSON
```

Explanation:

- Student and Teacher have common attributes like Name, Age, Address.
- These common attributes are stored in the higher-level entity **Person**.

2. Specialization

Definition:

Specialization is the process of dividing a higher-level entity into smaller lower-level entities based on their characteristics.

It follows a **top-down approach**.

Example:

```
      EMPLOYEE
                |
         ---------------
         |             |
      MANAGER       ENGINEER
```

Explanation:

- Employee is the main entity.
- Manager and Engineer are specialized entities.
- Each subclass has additional properties.

3. Aggregation

Definition:

Aggregation is a concept where a relationship between entities is treated as a higher-level entity.

It is used when a relationship needs to participate in another relationship.

Example:

```
EMPLOYEE -------- WORKS_ON -------- PROJECT
```

Here, **Works_On** relationship can be considered as an entity because it connects Employee and Project.

Difference Between Generalization and Specialization

Generalization	Specialization
Combines entities	Divides entities
Bottom-up approach	Top-down approach
Creates a super class	Creates subclasses
Example: Student + Teacher → Person	Employee → Manager, Engineer

Advantages of EER Model

1. Represents complex real-world objects.
2. Supports inheritance.
3. Reduces data redundancy.
4. Improves database design.
5. Provides better relationships between entities.

Conclusion

The EER model extends the ER model by adding concepts like generalization, specialization, and aggregation. It provides a powerful way to represent complex database systems.

9: Design an EER Diagram for a Hospital Management System Incorporating Inheritance and Weak Entities.

Explanation

A **Hospital Management System** manages information about patients, doctors, medical records, and treatments.

The EER model uses:

- **Inheritance** → To represent common features of doctors and patients.
- **Weak Entity** → To represent entities that depend on another entity for their existence.

Entities and Attributes

1. PERSON (Super Entity)

Stores common details.

Attributes:

- **Person_ID (Primary Key)**
- Name
- Address
- Phone

2. PATIENT (Sub Entity)

Inherited from Person.

Attributes:

- Patient_ID
- Disease

3. DOCTOR (Sub Entity)

Inherited from Person.

Attributes:

- Doctor_ID
- Specialization

4. MEDICAL_RECORD (Weak Entity)

Stores patient medical history.

Attributes:

- Record_ID (Partial Key)
- Date
- Diagnosis
- Treatment

EER Diagram

```
        PERSON
          --------------------
          Person_ID (PK)
          Name
          Address
          Phone
                |
          ----------------
          |              |
       PATIENT        DOCTOR
     -----------    ------------
```

```
   Patient_ID     Doctor_ID
   Disease        Specialization
          |
          |
          | 1:M
          |
MEDICAL_RECORD
----------------
Record_ID
Date
Diagnosis
Treatment
```

Concept Explanation

1. Inheritance

- PERSON is the superclass.
- PATIENT and DOCTOR are subclasses.
- Patient and Doctor inherit common attributes from Person.

Example:

- Person → Name, Address, Phone
- Patient → Disease
- Doctor → Specialization

2. Weak Entity

- Medical_Record is a weak entity.
- It cannot exist without a Patient.
- It depends on Patient's Patient_ID.

Relationships

1. **Person → Patient/Doctor**

    - One person can be either a patient or doctor.
2. **Patient → Medical_Record (1:M)**
    - One patient can have many medical records.
3. **Doctor → Patient (M:N)**
    - Doctors can treat many patients, and patients can consult many doctors.

Advantages of EER Design

- Reduces duplicate data.
- Provides better organization.
- Supports inheritance.
- Represents real-world hospital operations.

10: Explain the Role of a DBA (Database Administrator) and the Responsibilities Associated with the Position.

Database Administrator (DBA)

Definition:

A **Database Administrator (DBA)** is a person responsible for managing, maintaining, securing, and controlling the database system of an organization.

The DBA ensures that the database is available, reliable, and performs efficiently.

Roles and Responsibilities of DBA

1. Database Design and Creation

- Creates databases, tables, and relationships.
- Defines database structure according to requirements.

2. Installation and Configuration

- Installs and configures database management software.
- Maintains database environment.

3. Security Management

- Creates user accounts and passwords.

- Provides access permissions.
- Protects data from unauthorized access.

4. Backup and Recovery

- Performs regular database backups.
- Restores data after system failures or data loss.

5. Performance Monitoring

- Monitors database speed and performance.
- Improves query execution and database efficiency.

6. Data Integrity Maintenance

- Ensures accuracy and consistency of data.
- Applies constraints and validation rules.

7. User Management

- Creates and manages database users.
- Assigns different access privileges.

8. Database Maintenance

- Updates database software.
- Removes unnecessary data.
- Manages storage space.

9. Troubleshooting

- Identifies and solves database errors.
- Handles system failures.

10. Ensuring Availability

- Ensures the database is available whenever users need it.
- Maintains high reliability.

Conclusion

A DBA plays an important role in managing the database system. The DBA ensures **security, data integrity, backup, recovery, and efficient performance** of the database.